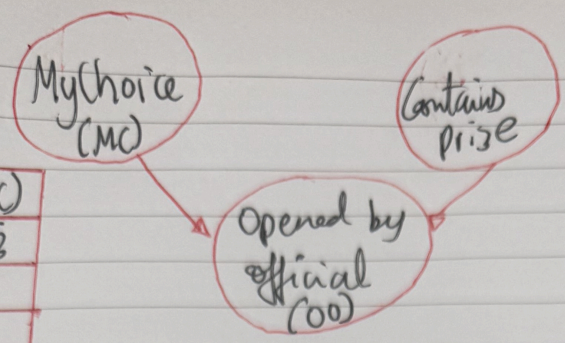

MC	P(MC)
A	$0.\overline{333}$
B	$0.\overline{333}$
C	$0.\overline{333}$

P	P(P)
A	$0.\overline{333}$
B	$0.\overline{333}$
C	$0.\overline{333}$

MC	A			B			C		
P / OO	A	B	C	A	B	C	A	B	C
A	0	0	0	0	0.5	1	0	1	0.5
B	0.5	0	1	0	0	0	1	0	0.5
C	0.5	1	0	1	0.5	0	0	0	0

$P(OO=B \mid MC=A, P=A) = 0.5 \cdot P(A) = 0.\overline{166}$

$P(OO=B \mid MC=A, P=B) = 0 \cdot P(B) = 0$

$P(OO=B \mid MC=A, P=C) = 1 \cdot P(C) = 0.\overline{333}$

$P(O=B \mid MC=A) = 0.\overline{1666} + 0 + 0.\overline{333} = 0.5$

To compute the posterior probability of the prize location after seeing that the official opened B, we use Bayes rule and normalize over all locations.

$P(P=A \mid MC=A, O=B) = \dfrac{0.\overline{166}}{0.5} = 0.\overline{333} \Rightarrow \boxed{\dfrac{1}{3}}$ } if you don't switch

$P(P=C \mid MC=A, O=B) = \dfrac{0.\overline{333}}{0.5} = 0.\overline{666} \Rightarrow \boxed{\dfrac{2}{3}}$ } if you switch

Formula: $P(P=p \mid MC=A, O=B) = \dfrac{P(O=B \mid MC=A, P=p) \cdot P(P=p)}{\sum_{p \in \{A,B,C\}} P(O=B \mid MC=A, P=p) \cdot P(P=p)}$